```
import pandas as pd
import numpy as np
```

```
df= pd.read_csv("/content/Business Retail.csv")
```

```
df.head()
```

	Order_ID	Customer_ID	Order_Date	Region	Product_Category	Customer_Segment	Quantity	Unit_Price	Discount_Rate	Revenue
0	1	CUST3818	8/18/2024	North	Clothing	Corporate	5	300.68	0.27	1097.48
1	2	CUST9689	6/19/2024	South	Beauty	Home Office	9	32.89	0.02	290.09
2	3	CUST9147	11/21/2024	West	Sports	Corporate	5	345.61	0.25	1296.04
3	4	CUST7938	7/19/2024	North	Clothing	Consumer	1	444.50	0.06	417.83
4	5	CUST5127	10/28/2024	South	Home & Kitchen	Consumer	5	65.13	0.21	257.26

Next steps: Generate code with df New interactive sheet

```
df.info()
df.describe()
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 13 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Order_ID          10000 non-null  int64
 1   Customer_ID       10000 non-null  object
 2   Order_Date        10000 non-null  object
 3   Region            10000 non-null  object
 4   Product_Category  10000 non-null  object
 5   Customer_Segment  10000 non-null  object
 6   Quantity          10000 non-null  int64
 7   Unit_Price        10000 non-null  float64
 8   Discount_Rate     10000 non-null  float64
 9   Revenue           10000 non-null  float64
 10  Cost              10000 non-null  float64
 11  Profit            10000 non-null  float64
 12  Payment_Method    10000 non-null  object
dtypes: float64(5), int64(2), object(6)
memory usage: 1015.8+ KB
```

	0
Order_ID	0
Customer_ID	0
Order_Date	0
Region	0
Product_Category	0
Customer_Segment	0
Quantity	0
Unit_Price	0
Discount_Rate	0
Revenue	0
Cost	0
Profit	0
Payment_Method	0

dtype: int64

```
df['Order_Date'] = pd.to_datetime(df['Order_Date'])
```

```
numeric_cols = ['Quantity', 'Unit_Price', 'Discount_Rate',
                'Revenue', 'Cost', 'Profit']

df[numeric_cols] = df[numeric_cols].apply(pd.to_numeric, errors='coerce')
```

```
df = df.drop_duplicates(subset='Order_ID')
```

```
df = df.dropna(subset=['Revenue', 'Cost', 'Profit'])
```

```
df['Discount_Rate'] = df['Discount_Rate'].fillna(0)
```

```
df['Calculated_Revenue'] = (
    df['Quantity'] * df['Unit_Price'] * (1 - df['Discount_Rate'])
)
```

```
df['Revenue_Difference'] = df['Revenue'] - df['Calculated_Revenue']

df['Revenue_Difference'].describe()
```

	Revenue_Difference
count	10000.000000
mean	-0.000020
std	0.002839
min	-0.005000
25%	-0.002400
50%	0.000000
75%	0.002400
max	0.005000

dtype: float64

```
df['Revenue'] = df['Calculated_Revenue']
```

```
df['Profit'] = df['Revenue'] - df['Cost']
```

```
df = df[df['Quantity'] > 0]
df = df[df['Unit_Price'] > 0]
df = df[df['Cost'] >= 0]
```

```
Q1 = df['Revenue'].quantile(0.25)
Q3 = df['Revenue'].quantile(0.75)
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

df = df[(df['Revenue'] >= lower) & (df['Revenue'] <= upper)]
```

```
df['Year'] = df['Order_Date'].dt.year
df['Month'] = df['Order_Date'].dt.month
df['Month_Name'] = df['Order_Date'].dt.month_name()
df['Year_Month'] = df['Order_Date'].dt.to_period('M')
```

```
def discount_category(x):
    if x == 0:
        return "No Discount"
    elif x <= 0.1:
        return "Low"
    elif x <= 0.3:
        return "Medium"
```

```
    else:
        return "High"

df['Discount_Category'] = df['Discount_Rate'].apply(discount_category)
```

```
text_cols = ['Region', 'Product_Category',
             'Customer_Segment', 'Payment_Method']

for col in text_cols:
    df[col] = df[col].str.strip().str.title()
```

```
df.info()
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 9847 entries, 0 to 9999
Data columns (total 20 columns):
```

```
df.to_csv("Cleaned_Business_Analytics_Dataset.csv", index=False)
```

```
 1   Customer_ID          9847 non-null   object
```

```
df.head()
```

```
 4   Product_Category     9847 non-null   object
 5   Customer_Segment     9847 non-null   object
 6   Quantity             9847 non-null   int64
 7   Unit_Price           9847 non-null   float64
```

1 to 5 of 5 entries Filter

index	Order_ID	Customer_ID	Order_Date	Region	Product_Category	Customer_Segment	Quantity	Unit_Price	Discount_Rate	Revenue
0	1	CUST3818	2024-08-18 00:00:00	North	Clothing	Corporate	5	300.68	0.27	1097.482
1	2	CUST9689	2024-06-19 00:00:00	South	Beauty	Home Office	9	32.89	0.02	290.08979999999997
2	3	CUST9147	2024-11-21 00:00:00	West	Sports	Corporate	5	345.61	0.25	1296.0375000000001
3	4	CUST7938	2024-07-19 00:00:00	North	Clothing	Consumer	1	444.5	0.06	417.83
4	5	CUST5127	2024-10-28 00:00:00	South	Home & Kitchen	Consumer	5	65.13	0.21	257.2635

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```
19  Discount_Category   9847 non-null   object
dtype: datetime64[ns](1), float64(7), int32(2), int64(2), object(7), period[M](1)
memory usage: 1.5+ MB
```

Next steps: Generate code with df New interactive sheet

```
                    0
Order_ID            0
```

```
from google.colab import files
files.download("Cleaned_Business_Analytics_Dataset.csv")
```

```
Region              0
Product_Category    0
Customer_Segment    0
Quantity            0
Unit_Price          0
Discount_Rate       0
Revenue             0
Cost                0
Profit              0
Payment_Method      0
Calculated_Revenue  0
Revenue_Difference  0
Year                0
Month               0
Month_Name          0
```